Sachin Mahajan

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science student who ships end-to-end ML systems, LLM-based applications, and backend APIs, including production RAG pipelines, browser extensions with real users, and scalable ML deployments running CI/CD.

EXPERIENCE

Podman — Red Hat / containers | Open-Source Contributor **Aug 2026 – Sep 2026**

- Fixed container storage auto-recovery in Podman by updating Quadlet service generator logic in Go, enabling automatic volume recreation on container restarts; added integration tests & documentation, merged into main (PR #29375)

TensorFlow – Google | Open-Source Contributor **Apr 2026 – Present**

- Fixed numerical instability in Exponential Linear Unit / Scaled Exponential Linear Unit (ELU/SELU) activations by replacing exp(x)-1 with expm1(x) in C++ (Eigen kernels), eliminating precision loss for small inputs; under review (PR #116947)
- Resolved underflow in tf.math.sign for complex values using stable magnitude computation (std::abs / hypot) in C++, preventing float32 underflow for values < 1.08e-19; approved by maintainers (PR #116986)

pandas — NumFOCUS / pandas-dev | Open-Source Contributor **Apr 2026**

- Fixed a Pandas Categorical.map bug where defaultdict-based mappings incorrectly replaced missing values with NaN by correcting missing-key handling logic; added regression tests and merged into the Pandas 3.1 milestone (PR #65413)

JobFit AI — Chrome & Edge Extension | Independent Project | Live (Chrome, Edge) **Apr 2026**

- Shipped a cross-browser extension (Chrome & Edge) using JavaScript and Groq LLM API for real-time resume-job fit scoring across LinkedIn & Indeed; migrated backend to Cloudflare Workers with Bring Your Own Key (BYOK) support.
- Designed privacy-first weighted scoring model (Skills 40%, Experience 35%, Education 15%) with local data storage.

SKILLS

Programming Languages and Core CS: Python, C++, Golang

Computer Fundamentals: Data Structures and Algorithms (DSA), Problem Solving, OOP, DBMS, Computer Networks

Machine Learning and Deep Learning: TensorFlow, Keras, scikit-learn, Embeddings, Transformers, Model Evaluation

LLMs and AI Engineering: LangChain (LCEL), LangGraph, CrewAI, Groq API, RAG, Vector Databases, ChromaDB

Backend and APIs: Django, FastAPI, REST APIs, PostgreSQL, MySQL, Gunicorn

DevOps and Tools: Docker, GitHub Actions, Git (Version Control), Open Source Contributions, Linux

PROJECTS

MediScan AI — Multi-Agent Medical Lab Report Intelligence | GitHub | Live **May 2026 – Jun 2026**

- Engineered a 7-node LangGraph pipeline combining ChromaDB RAG (164 chunks) and 6 clinical formulas (estimated Glomerular Filtration Rate (eGFR), Friedewald LDL) to generate explainable health reports from lab text or PDFs.
- Implemented a self-correcting CrewAI agent loop with Qwen3-32b as judge and conditional retry routing (max 2 iterations).
- Fixed a CrewAI/Groq cache_breakpoint bug via monkey-patch to unblock the agent loop in production.
- Scaled for production with MD5 caching (69s → under 1s), asyncio.Semaphore-limited Groq calls, /metrics endpoint, and 16-test GitHub Actions CI/CD; deployed via Docker on Hugging Face Spaces.

Tech Stack: Python, LangGraph, CrewAI, FastAPI, Groq, ChromaDB, PyMuPDF, fpdf2, Docker, GitHub Actions

Skill-Based Job Recommendation Engine (LLM-Assisted) | GitHub Link | Live Link **Dec 2025 – Jan 2026**

- Launched a full-stack job recommendation system using Django and PostgreSQL to map user skills to roles; devised set-based ranking and tuned ORM queries (prefetch_related) to efficiently handle 75+ jobs and 60+ skills.
- Integrated Groq LLM for resume parsing and skill gap analysis; implemented custom Django middleware for request latency tracking and ensured low-latency responses (<150 ms) through optimized API handling.
- Containerized the application using Docker and deployed on Render via CI/CD; designed RESTful APIs (recommendation, parsing, metrics) with structured responses and scalable backend architecture.

Tech Stack: Python, Django, PostgreSQL, Groq LLM, Docker, Gunicorn, GitHub Actions, REST APIs, HTML/CSS/JS

Phishing URL Detection | GitHub Link | Live Link **Oct 2025 – Nov 2025**

- Constructed a phishing URL detection system using 58 attribute features and XGBoost on 11,000+ URLs, achieving 91.38% accuracy; containerized with Docker and automated via GitHub Actions CI/CD pipeline.
- Shipped a Streamlit application for real-time URL prediction with 0.97 Receiver Operating Characteristic – Area Under Curve (ROC-AUC) and balanced precision-recall.

Tech Stack: Python, XGBoost, Scikit-learn, Pandas, NumPy, Streamlit, GitHub Actions, Docker

CERTIFICATES AND ACHIEVEMENTS

- HackerRank Intermediate SQL | Oracle Cloud Infrastructure AI Foundations | 400+ DSA (LC/GFG)

EDUCATION

Lovely Professional University, Jalandhar, Punjab **Aug 2023 – Jul 2027**

B.Tech in Computer Science and Engineering with Artificial Intelligence and Machine Learning **CGPA: 7.75**